METs PRACTICAL PROJECT

DATA ANALYST TEXTBOOK

Microsoft account
[Email address]

DATA ANALYSIS WORKFLOW MODULE

Course: Data Analyst Professional
Module Title: From Raw Data to Insights

🎯 Module Objective

By the end of this module, learners will:

- Understand the full data analysis lifecycle
- Work with real-world messy data
- Build a dashboard
- Present business insights

□ PART 1: THE WORKFLOW (TEACHING SECTION)

Step 1: Business Understanding

Scenario:

You are working for a retail company.

Business Problem:

"Sales have been fluctuating. Management wants to understand trends and improve performance."

Tasks:

- Define KPIs:
  - Total Sales
  - Profit
  - Sales by Region
  - Monthly Growth

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Step 2: Data Collection

Provide students with a dataset (you can create this in Excel):

Dataset Columns:

- Order_ID
- Order_Date
- Region
- Product
- Category
- Sales
- Cost
- Customer_ID

Step 3: Data Cleaning (Practical)

Common Issues to include in your dataset:

- Missing Sales values
- Duplicate rows
- Incorrect date formats
- Negative values (errors)

Student Tasks:

- Remove duplicates
- Handle missing values
- Fix date formats
- Validate numbers

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☞ Tool Options:

- Excel (Beginner)
- SQL / Python (Advanced)

Step 4: Data Transformation

Students must:

- Create new column:
  - Profit = Sales - Cost
- Extract:
  - Month from Order_Date
- Group data:
  - Total sales per region

Step 5: Exploratory Data Analysis (EDA)

Questions students must answer:

- Which region has the highest sales?
- Which product performs best?
- Are sales increasing over time?
- Any unusual spikes or drops?

Step 6: Data Analysis

Deeper Analysis:

- Profitability by category
- Monthly growth trends
- Top 10 customers

Step 7: Data Visualization (Power BI Project)

Students must build:

Dashboard Requirements:

- KPI Cards:
  - Total Sales
  - Total Profit
- Charts:
  - Sales by Region (Bar Chart)
  - Sales Trend (Line Chart)
  - Top Products (Bar Chart)

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Step 8: Insights & Recommendations

Students must answer:

1. What are the key findings?
2. What problems exist?
3. What actions should the company take?

Step 9: Presentation

Students present:

- Dashboard
- Insights
- Business recommendations

□ PART 2: FULL PRACTICAL PROJECT

🏢 Project Title:

Retail Sales Performance Analysis

💼 Project Brief

You are hired as a Data Analyst to analyze company sales data and provide insights to improve performance.

📌 Deliverables

Students must submit:

1. Cleaned dataset

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2. Excel/SQL work
3. Power BI Dashboard
4. Final report (PDF or presentation)

🛠□ Step-by-Step Student Guide

◆ Task 1: Load Data

- Import dataset into Excel or Power BI

◆ Task 2: Clean Data

Checklist:

- Remove duplicates
- Handle null values
- Fix formats

◆ Task 3: Transform Data

Create:

- Profit column
- Month column

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◆ Task 4: Analyze Data

Answer:

- Total revenue?
- Most profitable product?
- Weakest region?

◆ Task 5: Build Dashboard

Must include:

- Filters (Region, Product)
- At least 3 charts
- KPI indicators

◆ Task 6: Business Report

Writing

Retail Sales Analysis Report – METs Academy

1. Overview
This report analyzes sales performance across regions and products.

2. Key Findings

- Region X generated the highest revenue
- Product Y is the most profitable
- Sales declined in Month Z

3. Problems Identified

- Low performance in certain regions
- High costs affecting profit

4. Recommendations

- Increase marketing in low-performing regions
- Focus on high-profit products
- Reduce operational costs

5. Conclusion
Data-driven decisions can significantly improve business performance.

□📊 PART 3: ASSESSMENT (For Your Academy)

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✅ Marking Rubric (100 Marks)

- Data Cleaning: 20 marks
- Transformation: 15 marks
- Analysis: 20 marks
- Dashboard: 25 marks
- Insights & Presentation: 20 marks

🚀 BONUS: REAL-WORLD SKILLS YOU'RE TEACHING

This module builds:

- Analytical thinking
- Business understanding
- Technical skills (Excel, Power BI, SQL)
- Communication skills

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